

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 2.4: Acids, Bases and Salts — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.4: Acids, Bases and Salts
Driving Question	DRIVING QUESTION: Why do some substances from our everyday Kenyan environment — lake water, lemon juice, wood ash, cooking salt, antacids — react so differently with metals, food and living things, and how can understanding their acid, base or salt chemistry help us use them safely and sustainably?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Predict and Anchor: What Is Happening at Lake Magadi and Inside the Sufuria?	Students observed two striking photographs: Lake Magadi with white trona crusts and pink-red alkaline water where flamingos live, and a household aluminium sufuria with white mineral scale inside and acid-pitting outside from lemon-water left overnight. They also observed that 12 familiar Kenyan substances are difficult to classify without evidence.	Students have not yet learned any new content. They have surfaced what they already know (or believe they know) about acids, bases and salts from everyday Kenyan life and from previous grades. They have learned that their own prior knowledge produces disagreements that science can resolve.	Students produced their first explanatory model: a two-panel sketch of the Lake Magadi and sufuria phenomena labelled with their current best guesses. These explanations are deliberately tentative and incomplete and will be revised across Lessons 2 through 7 as evidence accumulates.
Lesson 2 pH Investigation: Measuring the Acid-Base Character of Everyday Kenyan Substances	Students tested 12 substances — including Lake Magadi water (simulated), lemon juice, wood-ash filtrate, cooking salt solution, vinegar, antacid solution, tap water, sugar solution, dilute HCl, dilute NaOH, milk, and baking soda solution — using universal indicator paper and a pH colour chart. They recorded indicator colour, pH value, and acid/base/neutral classification for each substance, and flagged surprises by comparing actual pH results against their Lesson 1 predictions.	Substances are classified as acids ($\text{pH} < 7$, dominant H^+ ions), bases ($\text{pH} > 7$, dominant OH^- ions), or neutral ($\text{pH} = 7$, equal H^+ and OH^-). Lake Magadi water is strongly basic ($\text{pH} 9\text{--}11$) because dissolved trona (sodium sesquicarbonate) releases OH^- ions — explaining why it is caustic to unprotected human skin. Lemon juice is acidic ($\text{pH} 2\text{--}3$) because citric acid releases H^+ ions — explaining why it pitted the aluminium sufuria overnight. Cooking salt (NaCl) solution is neutral ($\text{pH} 7$). Wood-ash solution is basic ($\text{pH} 9\text{--}11$). Evidence does not always match intuitive predictions — surprises are more scientifically valuable	Acids release hydrogen ions (H^+) when dissolved in water; the more H^+ ions, the lower the pH and the more acidic the substance. Bases release hydroxide ions (OH^-); the more OH^- ions, the higher the pH and the more basic the substance. The pH scale runs from 0 (most acidic) to 14 (most basic), with 7 as neutral. The corrosive behaviour of Lake Magadi water on unprotected skin and the pitting of the aluminium sufuria by lemon juice are both explained by excess ions (OH^- and H^+ respectively) reacting with biological or metallic surfaces. Salts such as NaCl can be neither acidic nor basic — a finding that

		than confirmations.	raises new questions about salt classification, to be explored from Lesson 3 onwards.
Lesson 3 Lesson 3: Biological and Chemical Roles of Acids and Bases — Digestion, Respiration and Bleaching	Students observed: (1) an antacid tablet neutralising indicator-coloured dilute HCl in real time, watching the pH rise from ~1–2 (red) toward neutral (yellow-green) with visible CO₂ effervescence; (2) a labelled diagram on Reading Card 1 showing pH values at each region of the human digestive tract; (3) the reversible equation $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3$ on Reading Card 2 in the context of Kenyan long-distance athletes and high-altitude running in the Rift Valley; (4) a comparison on Reading Card 3 between the pH of sodium hypochlorite bleach (~11–13) and Lake Magadi water (~10.5).	Students learned that: (1) the stomach produces HCl (pH 1–2), which activates digestive enzymes and kills pathogens in food such as nyama choma or githeri; (2) the pancreas secretes sodium hydrogen carbonate (NaHCO₃) into the duodenum to neutralise stomach acid, raising the pH to 7–8 for intestinal enzyme function; (3) CO₂ produced during cellular respiration dissolves in blood to form carbonic acid (H₂CO₃), and the blood buffer system maintains pH at ~7.4; (4) sodium hypochlorite (NaOCl) is an alkaline bleaching agent that destroys colour pigments by oxidation — its high pH is comparable to Lake Magadi's caustic water; (5) acids and bases are not simply hazardous chemicals but are precisely regulated agents essential for life processes.	Students explained: (1) why a flamingo wading in Lake Magadi's alkaline water (pH ~10.5) can live there because flamingos are adapted to it (this lesson does not explain the mechanism of that adaptation); (2) why the lemon juice (citric acid) that is safe to drink can simultaneously pit an aluminium sufuria overnight — the H⁺ ions from the acid react with the aluminium metal (acid–metal reactions are investigated experimentally in Lesson 4, using magnesium as a model metal for aluminium), a reaction that does not occur inside the body because the stomach lining is protected by mucus; (3) why blood must maintain a near-neutral pH of 7.4 — even though CO₂ from respiration continuously generates carbonic acid, the bicarbonate buffer system (NaHCO₃/H₂CO₃) prevents dangerous acidification; (4) how the soda ash (Na₂CO₃, produced from trona, Na₃H(CO₃)₂·2H₂O) harvested commercially from Lake Magadi functions as an alkaline industrial bleaching and cleaning agent — the same category of chemistry as the sodium hypochlorite used in Kenyan laundries and hospitals.
Lesson 4 Reactions of Acids with Bases, Carbonates and Metals — Evidence from the Sufuria and the Antacid	In Experiment 1 I observed the universal indicator colour change from purple to red as HCl was added to NaOH, and the test tube felt warm. In Experiment 2 I observed vigorous fizzing when H₂SO₄ met marble chips and damp litmus paper turned red at the mouth of the flask. In Experiment 3 I observed magnesium ribbon disappearing in HCl with rapid fizzing and a squeaky pop from the glowing splint test.	I learned that acids react with bases in neutralisation to produce a salt and water; acids react with carbonates to produce a salt, water and carbon dioxide; acids react with metals to produce a salt and hydrogen gas. A salt is one of the products in each of these three acid reactions. The salt product depends on which acid and which base or metal react together.	These reactions explain why the aluminium sufuria was pitted overnight — the citric acid in lemon juice reacts with aluminium in a similar way to how HCl reacted with magnesium (our model metal), forming a salt and hydrogen gas and corroding the surface. They also explain why antacid tablets fizz in the stomach — the carbonate in the tablet reacts with stomach HCl to form a salt, water and CO₂. The mineral crusts at Lake Magadi are salts too, but they form by crystallisation as lake water evaporates, not by acid reactions.
Lesson 5 Salts — Classification	We observed four salts — NaCl, anhydrous CaCl₂, Na₂CO₃·10H₂O and CuSO₄·5H₂O	We learned that salts differ in how they interact with water vapour in the air. A	We explained that the canteen cook's salt in Mombasa clumps because NaCl is only

and Behaviour in Air: Why Does the Canteen Salt Clump in Mombasa Rains?	— left in open petri dishes for 24 hours. We measured and recorded mass changes and changes in visual appearance. We found that CaCl_2 gained mass and partially dissolved into a liquid, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ lost mass and turned powdery, NaCl showed very little change, and $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ showed slight mass loss.	hygroscopic salt absorbs moisture from the air without dissolving. A deliquescent salt absorbs so much water vapour that it dissolves completely. An efflorescent salt loses its water of crystallisation to the air and becomes a dry powder. These differences depend on how strongly the ions in the salt lattice attract water molecules compared to how strongly the atmosphere holds water vapour.	weakly hygroscopic — our dish showed very little change, but over weeks of very high coastal humidity it absorbs enough moisture to clump. We explained that the Lake Magadi trona crusts can shrink and crack in the wet season because $\text{Na}_3\text{H}(\text{CO}_3)_2 \cdot 2\text{H}_2\text{O}$ is a hydrated salt whose crust gains or loses water with humidity; the crusts themselves form by crystallisation as lake water evaporates, not by efflorescence. We connected the water of crystallisation in hydrated salt formulas (such as $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ and $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$) to the observable mass changes we measured, showing that the dot-number in a hydrated formula represents real water molecules held inside the ionic lattice that can be lost or gained.
Lesson 6 Applications of Salts in Real Life — From Lake Magadi to Your Kitchen, Farm and Clinic	Learners researched and mapped five real-life sectors (agriculture — CAN/DAP on Rift Valley farms; food industry — NaCl preservation of nyama choma and fish; medicine — ORS, saline, antacid tablets; laundry — washing soda $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$; road use and environment) in which specific named salts play essential roles, and they peer-taught these findings to mixed teaching groups using completed sector maps as visual aids.	Salts are not just a single substance (table salt / NaCl) but a vast family of ionic compounds produced by neutralisation reactions; each salt's specific properties — solubility, reactivity, hygroscopicity, deliquescence — determine which sector it is used in. Lake Magadi's trona (sodium sesquicarbonate) is the source of industrial soda ash (Na_2CO_3), the anhydrous form of domestic washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$), directly linking the anchoring phenomenon to everyday Kenyan laundry practice. Excess dietary NaCl disrupts osmotic balance and raises blood pressure, illustrating that the same salt that is essential for life can be harmful in excess — echoing the phenomenon's central puzzle of 'caustic yet useful.'	The chemical identity of a salt — its formula, ionic composition, and physical behaviour (including air-behaviour classification from Lesson 5) — determines its real-life application: CAN and DAP release ammonium and nitrate ions that plants need for protein synthesis; antacids contain basic salts (CaCO_3, $\text{Mg}(\text{OH})_2$) that neutralise excess stomach HCl; washing soda's high carbonate ion concentration softens hard water by precipitating Ca^{2+} and Mg^{2+} ions; and NaCl gritting on cold highland roads lowers the freezing point of water by disrupting the ionic lattice of ice — all explained by the same ionic bonding and dissolution principles studied in Lessons 1–5.
Lesson 7 Driving Question Board Synthesis and Cumulative Model Building	Students observed the full set of Driving Question Board entries accumulated since Lesson 1, peer models from all groups during the gallery walk, and the live consensus model built collaboratively on the board at the end of the lesson. They also observed their own Lesson 1 written predictions compared to their current understanding.	Students learned that all six lessons of evidence converge on a coherent explanatory framework: the pH scale, acid-base-salt definitions, three reaction types, three salt-air behaviours and real-life applications all connect to the two anchoring images. They learned that scientific models are revised iteratively as evidence accumulates, and that Lake Magadi trona ($\text{Na}_3\text{H}(\text{CO}_3)_2 \cdot 2\text{H}_2\text{O}$) is a	Students explained: why Lake Magadi water is caustic (high pH, high OH^- concentration from carbonate and bicarbonate salts) yet yields commercially and domestically useful soda ash; why lemon juice pitted the aluminium sufuria overnight (acid-metal reaction producing aluminium salt and hydrogen gas) but is safe to consume in dilute form; why the sufuria scale formed (dissolved mineral

		sodium carbonate salt and the raw material for the washing soda in domestic laundry, whereas the sufuria scale is a different salt, calcium carbonate — illustrating that one phenomenon can be explained by multiple intersecting chemistry concepts.	salts in hard water deposited on evaporation — salt-air and solubility behaviour); and how every phenomenon in the unit is an instance of the same underlying chemistry of acids, bases and salts acting on materials and living systems in the Kenyan environment.
Lesson 8 Final Explanation: Environmental Impact, Community Project Showcase & Assessment — Closing the Loop on Lake Magadi and the Sufuria	Students observed: (1) a case study with real seasonal nitrate concentration data from three Lake Victoria monitoring stations (Kisumu Port, Dunga Beach, Kendu Bay), showing higher values in the rainy season due to fertiliser run-off; (2) their own original Lesson 1 prediction sheets and conceptual model drawings, revealing how their understanding has grown; (3) the two anchoring phenomenon images (Lake Magadi soda crusts; pitted aluminium sufuria) displayed again for final explanation; (4) peer community sensitisation presentations (posters, dramas, pamphlets) on household acid-base chemistry and correct salt use.	Students learned: (1) excess fertiliser salts (CAN, DAP) dissolve in water, run off into Lake Victoria, cause eutrophication (algal bloom → oxygen depletion → fish death) — directly linking salt solubility and ion chemistry to environmental harm; (2) trona (sodium sesquicarbonate, $\text{Na}_3\text{H}(\text{CO}_3)_2 \cdot 2\text{H}_2\text{O}$, the source of Na_2CO_3 soda ash) at Lake Magadi is an alkaline salt that crystallises as lake water evaporates in the Rift Valley heat; (3) lemon juice (citric acid, low pH) reacts with aluminium metal ($\text{Al} + \text{acid} \rightarrow \text{aluminium salt} + \text{H}_2$), pitting the sufuria; (4) canteen salt clumps in April rains because NaCl is hygroscopic — Na^+ and Cl^- ions attract polar water molecules from humid air; (5) their own conceptual models changed substantially across eight lessons, and scientific understanding grows by revising prior models with new evidence.	Students explained, using the Claim–Evidence–Reasoning framework: (1) WHY Lake Magadi's trona forms white shoreline crusts — evaporation concentrates Na_2CO_3 dissolved in the alkaline lake water, which then crystallises as the water dries; the high pH makes the water caustic yet commercially valuable as soda ash for the Magadi Soda Company; (2) WHY lemon juice pitted the aluminium sufuria — citric acid (H^+ ions, pH 2–3) reacted with the aluminium metal surface overnight via an acid-metal reaction, producing aluminium salt and hydrogen gas, corroding the pot wall; (3) WHY canteen salt clumps in the long rains — sodium chloride is weakly hygroscopic, absorbing small amounts of water vapour from humid April air onto its ionic crystal surfaces; (4) HOW fertiliser salt overuse harms Lake Victoria — nitrate and phosphate ions from CAN and DAP run off into the lake, fuel explosive algal and water hyacinth growth, decomposing biomass depletes dissolved oxygen, and Nile tilapia (ngege) suffocate — all traceable to the solubility and ionic nature of salts studied throughout the sub-strand.

ATTRIBUTION AND LICENSING

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